ANA 222 Overview of lower limb, Femoral triangle, femoral sheath and canal

By Dr. Mbaka G.O.

Overview of lower limb:

The lower limb is essentially similar to the upper limb as both in most animals are used for body movement.

- Each basically has a girdle- Upper and lower limb girdles.
- The girdle support three segments of the limb; proximal thigh or arm, middle leg or forearm and distal foot or hand.
- They both have similar bone number at each segment.

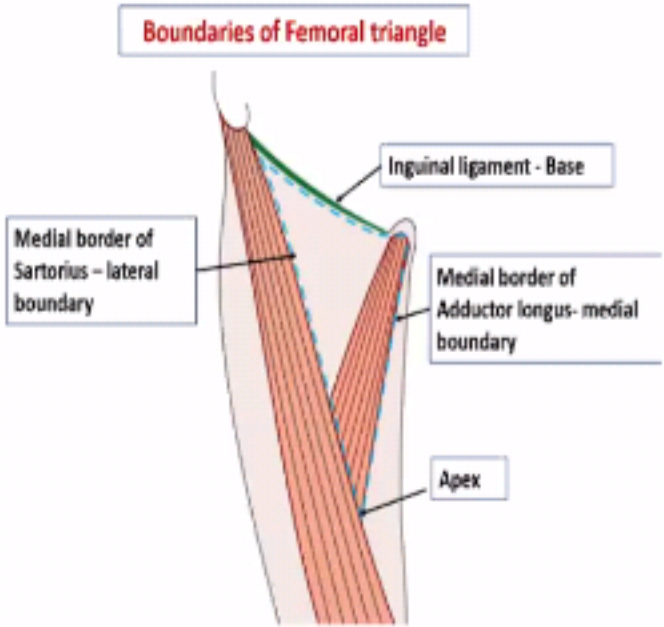
- The muscles are similar but the actions they produce are different because of rotational difference of the developing limb buds.
- In humans due to the evolution of erect posture, not withstanding their basic similarities because of their functional requirement have made rotational variations to adapt to their new role.
- Consequently, the upper limb is specialized for apprehension with greater mobility.
- The lower limb is built for support and greater stability.
- Muscles of the lower limbs are much better developed to carry the body weight and providing stability during propulsion. The major muscles for propulsion being the gluteal muscles.

Developmental changes of both limbs

At the early stage of development, the lower limb bud rotate medially through 90° so that preaxial (tibial border) faces medial hence extensor compartment becomes anterior.

The upper limb also undergo 90° rotation in this case their preaxial (radial) border faces laterally. The extensor surfaces posteriorly.

- The triangle is an intermuscular space in the proximal part of anterior compartment of thigh.
- The sides and floor of the triangle is formed by muscles.
- Roof by fascia while the base by inguinal ligament.
- The lateral boundary is formed by the medial border of Sartorius
- The medial boundary is formed by medial border of adductor longus.

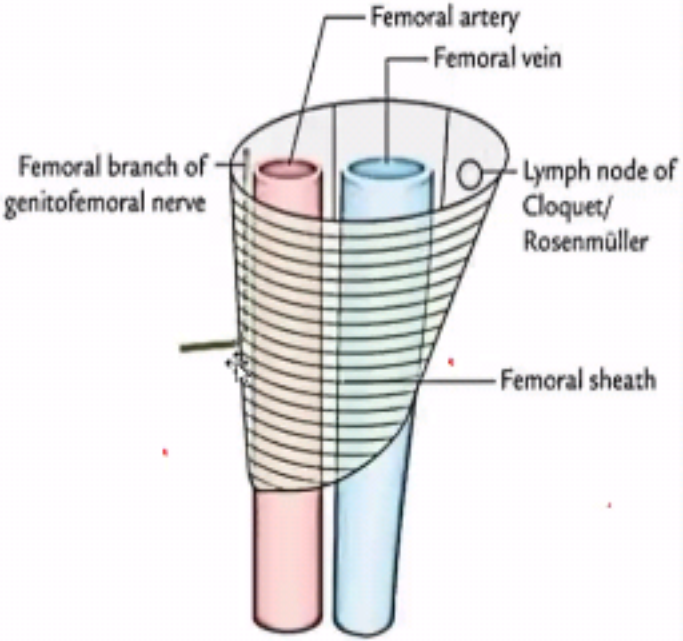


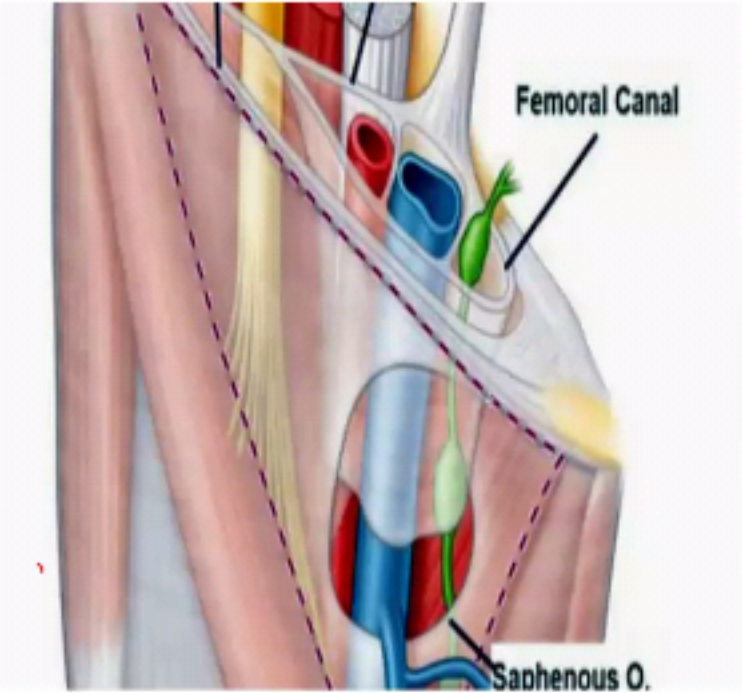
- Apex is formed where the Sartorius crosses the adductor longus.
- The base is formed by inguinal ligament.
- The floor is formed by the following muscles; iliacus, tendon of psoas, pectineus, and adductor longus.
- The roof is formed by superficial fascia containing superficial inguinal lymph nodes, the great saphenous vein, the deep fascia (fascia latae) pierced by saphenous vein at the saphenous opening.

- Contents: (i) Femoral artery from mid inguinal point to apex with 6 branches, 3 superficial & 3 deep branches. (ii) Femoral vein that lies medial to the artery and later posterior medial to it; receives tributaries from great saphenous vein, circumflex vein and corresponding veins to their arterial counterparts. (iii) Femoral sheath (iv) Femoral nerve (v) Pectineus nerve (vi) Femoral branch of genitofemoral nerve. (vii) Lateral cutaneous nerve (viii) Deep inguinal nodes for glans penis.

- Femoral sheath: Funnel shaped sheath of fascia enclosing upper 3-4 cm of femoral vessels.
- Formed by downward extension of 2 layers of fascia from abdominal wall.
- From anterior wall by fascia transversalis & from posterior wall by fascia iliacus.
- The medial and lateral walls are asymmetrical,
- While wall is oblique, the lateral wall is straight.
- The sheath is divided into 3 compartments; most lateral for artery, middle for vein and most medial for femoral canal.

Diagram of femoral sheath and femoral canal



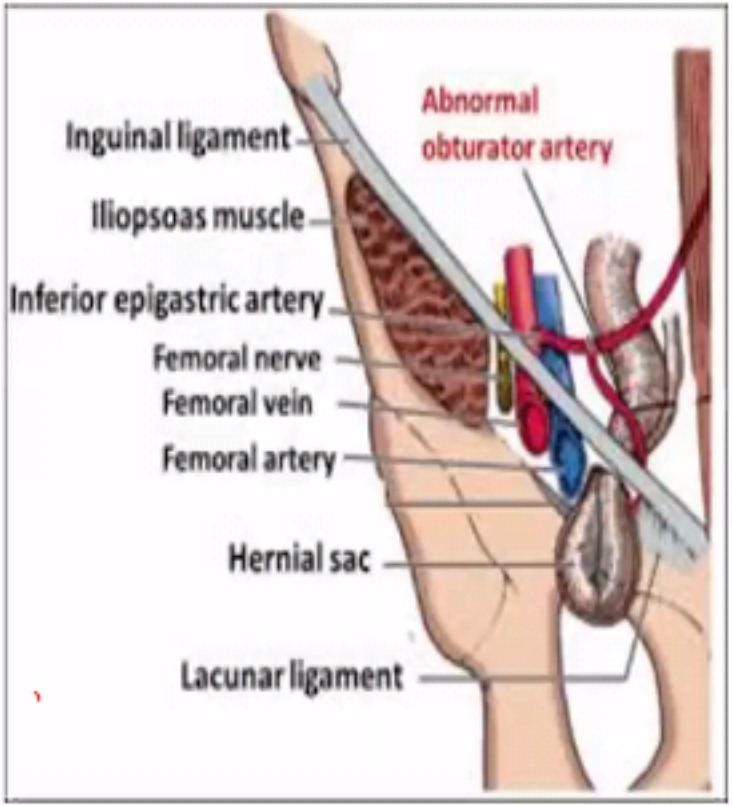


- Femoral canal: Situated medial to femoral vein.
- Conically shaped with base superior and apex inferior.
- It is 1.5cm in length and 1.5 cm in width at its widest part which is the base.
- The upper end of the canal, the base is called the femoral ring.
- The ring has four boundaries; Anteriorly is inguinal ligament, posteriorly, pectineus muscle & its facia enclosure.
- Medially is lacunar ligament and laterally is the septum separation between it and the femoral vein.
- The ring is enclosed by extraperitoneal fascia called femoral septum.
- The parietal peritoneum covering the septum shows depression called femoral fossa.
- Content: Cloquet lymph nodes that drain glans penis or clitoris, areolar tissue and lymphatic vessels.

- Femoral hernia: An area of potential weakness on abdominal wall through which abdominal content could bulge called femoral hernia.
- More common in females because the ring is larger in females due to larger size of pelvis while the vessels are smaller.
- It has neck and sac of abdominal bowel.
- The sac enters the canal through femoral ring, goes anteriorly and comes out through saphenous opening & upwards along superficial epigastric and superficial circumflex vessels.
- During reversal it is prone to obstruction at the ring which is rigid.
- Therefore concave lateral margin of lacunar ligament has to be cut.

- Caution has to be taken because abnormal obturator artery might lie on the pubic surface of lacunar ligament.

- The artery being formed by pubic branch of inferior epigastric that anastomosis with pubic branch of obturator artery. This might be a large vessel.



- Adductor canal: An intermuscular space on the medial side, middle ⅓ of thigh.
- It is triangular in cross section. It was a site used for ligation of femoral artery for the treatment of popliteal artery aneurysm.
- It extends from the apex of the femoral triangle to the tendinous opening on adductor magnus at its distal limit.
- It has the following boundaries; anteriorly, by vastus medialis; posteriorly, adductor longus at proximal part and adductor magnus at distal part. The medial wall which also represent the roof is formed by fibrous membranous sheath linking the vastus medialis with the two adductor muscles. On the sheath lies the sartorius.

Contents: (1) The femoral artery situated anterior to the vein. It gives off muscular branches and descending genicular branch. (2) femoral vein posterior lateral to the artery. (3) Saphenous nerve that crosses the artery anteriorly. (4) nerve to vastus medialis (5) Adductor nerves anterior and posterior divisions.

